DIPLOMA PROGRAMME PRACTICE · HL · SET A

Biology / Paper 1A

Question booklet · 40 marks · 120 minutes across 1A and 1B

Answer every question. Choose one option, A–D, per question. Use a calculator. Record choices online or on separate paper.

QUESTION 1 [1]

Habitability models compare two otherwise identical rocky planets. Which change most directly favours persistence of liquid surface water over geological time?

A Very low surface gravity combined with rapid atmospheric escape
B High mean surface temperature regardless of atmospheric pressure
C Sufficient gravity to retain an atmosphere and temperatures compatible with liquid water
D Maximum incident radiation irrespective of orbital distance or atmosphere

QUESTION 2 [1]

A double-stranded DNA fragment contains 180 base pairs, including 54 G–C pairs. How many thymine nucleotides does it contain?

A 72
B 54
C 252
D 126

QUESTION 3 [1]

In a DNA molecule with conventional antiparallel strands, one strand ends in a free 3′ hydroxyl group. To which position is the next nucleotide covalently attached during synthesis?

A The incoming nucleotide attaches to the template’s 3′ hydroxyl
B The 3′ hydroxyl of the growing strand joins the incoming nucleotide’s phosphate
C Two nitrogenous bases form a peptide bond
D The 5′ phosphate of the growing strand joins the incoming base

QUESTION 4 [1]

Simple amphipathic molecules form vesicles enclosing RNA in water. Which conclusion does this observation support?

A A membrane-bound vesicle alone demonstrates self-sustaining evolution
B A selectively enclosed compartment can arise before modern cells
C Compartment formation establishes that DNA preceded any catalytic RNA
D All early membranes required modern protein pumps to form

QUESTION 5 [1]

After a bacteriophage infects a cell, phage DNA integrates into the bacterial chromosome and copies during ordinary cell division. Which event would distinguish a subsequent lytic phase?

A Assembly of progeny phages followed by release through host lysis
B Inheritance of a repressed prophage by two daughter cells
C Repeated division of a host carrying an integrated prophage
D Maintenance of one prophage sequence per host chromosome

QUESTION 6 [1]

Two isolated bird populations look alike. Which observation most directly supports treating them as separate biological species?

A Their mean adult masses differ by 3%
B They have different frequencies of a plumage allele
C They occupy forests on different islands
D When brought into contact they repeatedly fail to produce fertile offspring despite normal breeding conditions

QUESTION 7 [1]

At equivalent positions in a conserved gene, species X and Y share six derived bases absent from Z and an outgroup. Other evidence is unavailable. Which tree is provisionally favoured?

A X and Y share a more recent common ancestor with each other than with Z
B The branch lengths from X and Y must be identical in real time
C Z is closer to X because Z retains ancestral bases
D X and Y share a derived state only when both also match the outgroup

QUESTION 8 [1]

Two fertile plant populations differ in chromosome number. A new individual with doubled chromosome sets can reproduce with similar polyploids but rarely with the original population. What may result?

A Loss of all homologous pairs from polyploid cells
B Conversion of all individuals to haploids
C Rapid reproductive isolation within the same geographic area
D Compulsory geographic isolation before any reproductive barrier

QUESTION 9 [1]

An island breeding programme can house only 20 founders. Which selection best protects genetic diversity across the source population?

A Use ten sibling pairs that already breed successfully
B Sample minimally related individuals across the source range
C Choose the twenty heaviest adults from one family
D Choose individuals expressing one preferred colour

QUESTION 10 [1]

After heating, a protein loses activity. Chemical analysis detects no change in amino acid sequence. Which conclusion is justified?

A The active site must remain functional as long as peptide bonds remain intact
B A changed side-chain sequence is the only way to alter active-site shape
C The functional three-dimensional conformation may have changed
D Loss of activity demonstrates cleavage of the polypeptide backbone into separate chains

QUESTION 11 [1]

An enzyme normally buries a nonpolar side chain in its interior. What may happen when mutation replaces it with a charged side chain?

A A substitution outside the active site cannot alter substrate binding
B A charged group in the buried core must stabilize folding regardless of nearby residues
C Only quaternary structure can change because the number of amino acids stays constant
D Folding stability and hence active-site shape may change even if the residue is distant from the active site

QUESTION 12 [1]

Two animal membrane samples contain equal phospholipids but different cholesterol concentrations. Which prediction reflects cholesterol's buffering role?

A Its effect is to increase fluidity by the same amount at every temperature
B At low temperatures it increases intermolecular packing by making the tails fully parallel
C At low temperatures it can impede tight phospholipid packing
D At high temperatures it increases phospholipid movement without restriction

QUESTION 13 [1]

A gland cell secretes a polypeptide enzyme. Which route would a newly made enzyme normally follow?

A Golgi apparatus → nucleus → rough ER
B Nucleus → lysosome → chloroplast
C Smooth ER → mitochondrion → nucleus
D Rough ER → Golgi apparatus → secretory vesicle

QUESTION 14 [1]

An electron micrograph shows two mitochondria of equal volume but different crista surface areas. Which inference is justified, assuming equal respiratory-protein density?

A Cristae provide extra sites for glycolysis in the matrix
B Surface area proves a higher measured ATP production rate under all conditions
C The one with more crista membrane could accommodate more electron-transfer and ATP synthase complexes
D The one with more cristae must replicate DNA faster

QUESTION 15 [1]

Fibrosis doubles the diffusion distance across a gas exchange surface. All other factors remain fixed. How does the initial oxygen diffusion rate change under an inverse-distance model?

A It is unchanged
B It doubles
C It falls to one quarter
D It halves

QUESTION 16 [1]

At the same oxygen partial pressure, adult haemoglobin shows lower saturation in actively respiring muscle than in resting muscle. Which local change contributes?

A Higher CO_2 converts haemoglobin into myoglobin
B A lower pH prevents oxygen dissociation
C Higher CO_2 and lower pH reduce oxygen affinity
D Lower CO_2 and higher pH reduce oxygen affinity

QUESTION 17 [1]

A ventricular pressure trace rises above atrial pressure but remains below arterial pressure. During early ventricular contraction, what is the valve configuration?

A Both sets of valves are open
B Only atrioventricular valves are open
C Only semilunar valves are open
D Both atrioventricular and semilunar valves are closed

QUESTION 18 [1]

A sarcomere shortens without changing thick filament length. Which combination is expected?

A The A band disappears while actin length doubles
B Z lines separate and the A band shortens
C Myosin filaments shorten and the I band widens
D Z lines approach and the A band remains approximately constant

QUESTION 19 [1]

Two desert plants have equal leaf area. Plant X has stomata recessed into still-air pockets. Which consequence is most likely in dry wind?

A A more humid local boundary layer reduces the water-vapour gradient
B X no longer requires xylem
C Water leaves faster because the diffusion path shortens
D X can take up carbon dioxide only at night

QUESTION 20 [1]

A substrate disappears from 8.0 to 6.2 mmol in 30 s. A matched tube lacking enzyme loses 0.30 mmol. What is the background-corrected mean enzymatic rate?

A 0.070 mmol s^{-1}
B 0.010 mmol s^{-1}
C 0.060 mmol s^{-1}
D 0.050 mmol s^{-1}

QUESTION 21 [1]

A competitive inhibitor reduces initial reaction rate. Which result best supports competition at the active site?

A At a sufficiently high substrate concentration, the inhibited rate approaches the original maximum
B Inhibition increases as the substrate-to-inhibitor ratio rises at constant enzyme concentration
C The maximum rate remains lower even when active sites are saturated with substrate
D The rate declines equally at every substrate concentration because enzyme molecules are permanently lost

QUESTION 22 [1]

Germinating seeds warm an insulated flask while an equal mass of boiled seeds does not. Which explanation best connects the observation to respiration?

A Oxygen supplies the carbon atoms of respiratory substrates
B All energy from substrates is stored indefinitely as ATP
C Some chemical energy released from substrates is transferred as heat
D Heat is produced only by photosynthesis

QUESTION 23 [1]

A chemical increases proton permeability of the mitochondrial inner membrane while electron transfer continues. What immediate change is most likely?

A All oxygen uptake must stop immediately
B Less ATP is generated per unit of oxygen consumed
C More ATP is generated because protons bypass ATP synthase
D The proton gradient increases because ions diffuse freely

QUESTION 24 [1]

Illuminated isolated thylakoids receive water, ADP, phosphate and $NADP^+$. Which pair is produced directly by the light-dependent reactions?

A Glucose and RuBP
B Pyruvate and acetyl-CoA
C Starch and carbon dioxide
D ATP and reduced NADP

QUESTION 25 [1]

A peptide hormone remains outside its target cell, yet intracellular enzyme activity changes rapidly. What is the best explanation?

A Binding a membrane receptor initiates intracellular signal transduction
B A change in enzyme activity shows that the peptide bound the enzyme active site directly
C The peptide must cross the bilayer and bind DNA before any response
D Rapid enzyme activation requires new transcription of the structural gene for that enzyme

QUESTION 26 [1]

A sensory neuron carries an impulse from a receptor toward the spinal cord. Which change moves along its axon?

A Diffusion of an action potential through blood
B A wave of membrane depolarization
C Movement of the receptor cell to the spinal cord
D Bulk flow of neurotransmitter inside the axon

QUESTION 27 [1]

A drug blocks voltage-gated Na^+ channels at the next node of Ranvier. What is most directly impaired?

A Regeneration of the action potential at that node
B Binding of all transmitters at every synapse
C Passive current beneath every myelin segment ceases by definition
D Synthesis of myelin lipid by the axon

QUESTION 28 [1]

A shoot bends toward unilateral light. Which observation best supports an auxin redistribution model?

A Before curvature, the shaded side shows higher auxin and faster cell elongation
B Growth stops permanently when the shoot is illuminated
C All cells on both sides elongate equally throughout the experiment
D Roots and shoots always show identical responses to auxin concentration

QUESTION 29 [1]

A closed population of beetles has 80 marked adults. A later unbiased sample contains 12 marked and 48 unmarked adults. What population estimate follows from equal capture probabilities and thorough mixing?

A 100
B 400
C 480
D 320

QUESTION 30 [1]

Both parental DNA strands carry a heavy label. Replication occurs twice in light medium. What fraction of double-stranded molecules still contains a heavy strand?

A None of them
B One quarter
C One half
D All of them

QUESTION 31 [1]

During PCR, amplification is initially ideal. Starting from 15 double-stranded target molecules, how many target molecules are present after six complete doubling cycles?

A 1920
B 960
C 90
D 480

QUESTION 32 [1]

A mutation eliminates a splice site in a eukaryotic gene. Which result most directly tests defective RNA processing?

A A change in intron retention in mature transcripts from the mutant
B A change in the number of chromosomes in every cell
C An increase in all tRNA anticodon lengths
D Failure of DNA to contain uracil

QUESTION 33 [1]

Cas9 editing yields the intended small deletion in 65% of cultured cells. What additional test addresses sequence specificity most directly?

A Measure the colour of the culture medium
B Increase the incubation temperature without further measurements
C Sequence plausible off-target genomic sites in edited and control cells
D Count only the edited cells that survive

QUESTION 34 [1]

A diploid cell contains 12 chromosomes in G1. Immediately after DNA replication and before mitosis, how many chromosomes and chromatids are present?

A 6 chromosomes and 12 chromatids
B 12 chromosomes and 24 chromatids
C 24 chromosomes and 24 chromatids
D 12 chromosomes and 12 chromatids

QUESTION 35 [1]

Two tissues from one individual contain the same gene sequence but different amounts of its mRNA. Which observation supports transcriptional regulation?

A More ribosomes bind each mRNA in one tissue with no change in transcription
B Different promoter occupancy by transcription factors in the two tissues
C The mature transcript is more stable while initiation is unchanged
D The mature protein is degraded more slowly while transcript initiation remains unchanged

QUESTION 36 [1]

Use water potential = solute potential + pressure potential. A cell has solute potential −0.75 MPa and pressure potential +0.25 MPa. In an external solution at −0.65 MPa, what is the initial net movement?

A Into the cell
B Into the compartment with the higher water potential
C No net movement because the cell is turgid
D Out of the cell

QUESTION 37 [1]

A heterozygote with a dominant phenotype is crossed with a homozygous recessive individual. What is the probability that two successive offspring both show the recessive phenotype?

A ¾
B ⅛
C ¼
D ½

QUESTION 38 [1]

A heterozygote AB/ab is test-crossed with ab/ab. Progeny counts are 410 AB, 390 ab, 95 Ab and 105 aB. What is the observed recombinant percentage?

A 40%
B 20%
C 80%
D 10%

QUESTION 39 [1]

Dehydration increases plasma osmolarity. In a healthy mammal, what response helps correct it?

A Increased ADH increases collecting-duct water permeability
B Increased ADH closes all aquaporins
C Decreased ADH increases collecting-duct water permeability
D The kidney returns less water to blood when ADH rises

QUESTION 40 [1]

At a locus with two alleles in Hardy–Weinberg equilibrium, the recessive phenotype frequency is 0.09. What is the expected heterozygote frequency?

A 0.42
B 0.21
C 0.30
D 0.49